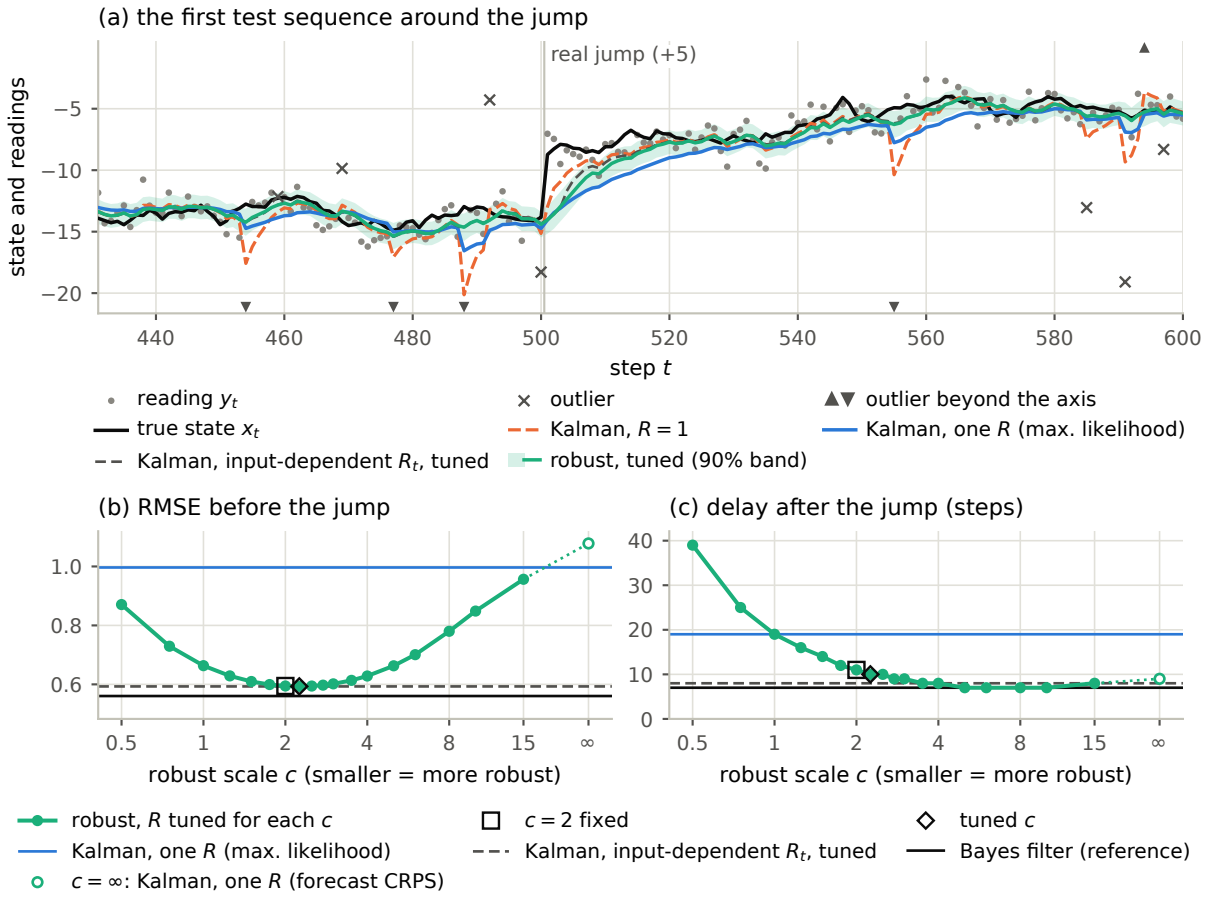
(a) the first test sequence around the jump
real jump (+5)
state and readings
step t
reading y_t
outlier
outlier beyond the axis
true state x_t
Kalman, $R = 1$
Kalman, one R (max. likelihood)
Kalman, input-dependent R_t, tuned
robust, tuned (90% band)
(b) RMSE before the jump
(c) delay after the jump (steps)
robust scale c (smaller = more robust)
robust scale c (smaller = more robust)
robust, R tuned for each c
$c = 2$ fixed
tuned c
Kalman, one R (max. likelihood)
Kalman, input-dependent R_t, tuned
Bayes filter (reference)
$c = \infty$: Kalman, one R (forecast CRPS)